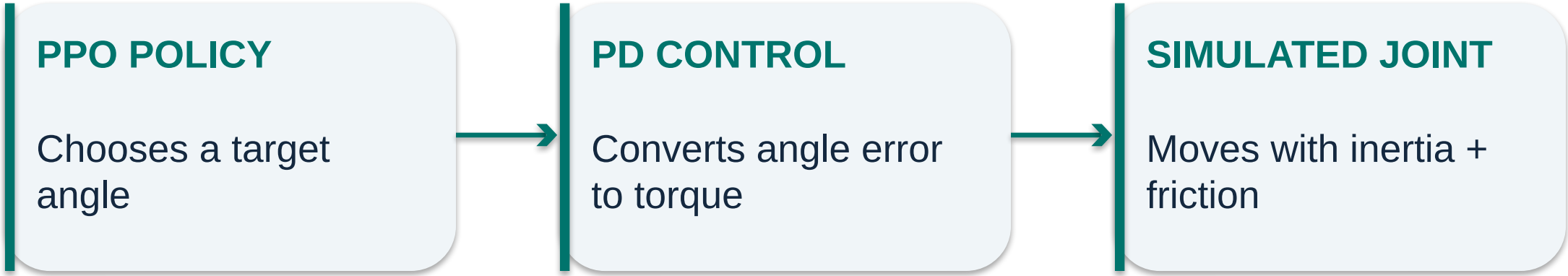
What am I demonstrating?

Can training with varied physics make a joint controller more robust?



Compare two PPO agents: fixed training physics versus randomized training physics.

SAY “Can training with varied physics help a joint controller handle an unfamiliar environment?”

Run the code and follow the workflow

Run the program from the working project folder:

```
.\.venv\Scripts\python.exe reading5_sim2real_ppo.py
```

1 TRAIN

Same PPO recipe.

Different training physics.

2 TEST

Nominal + held-out.

30 matched trials each.

3 VIEW

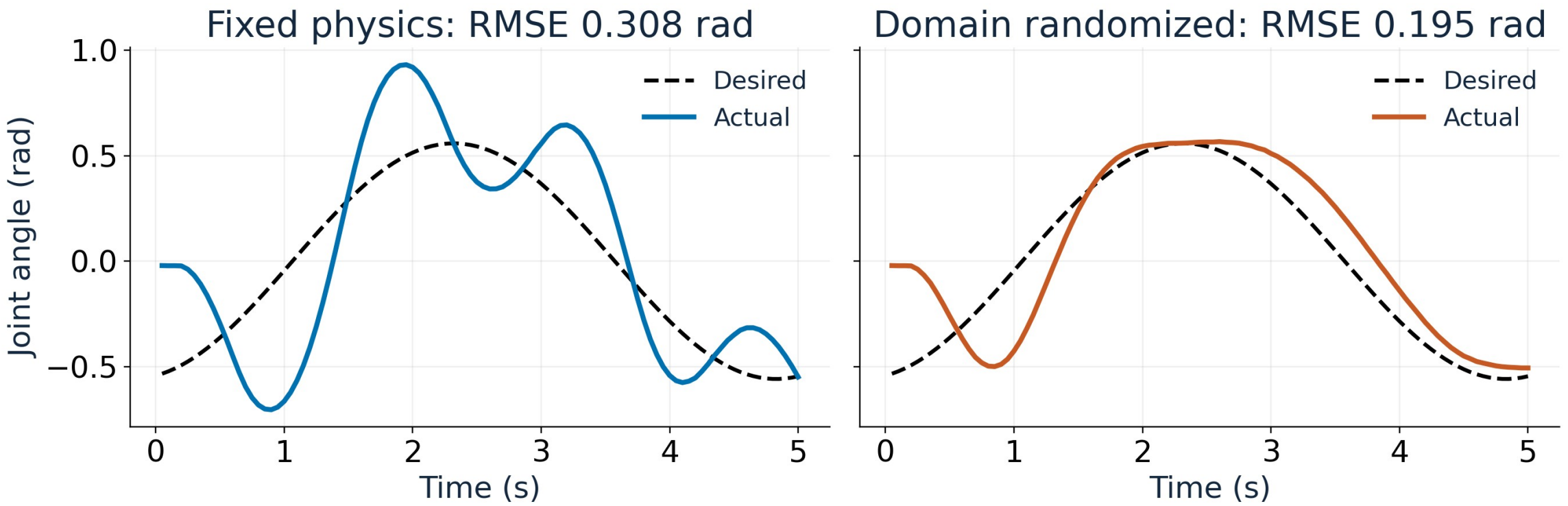
Print evaluation metrics.

Save three PNG graphs.

Code tour: Joint.step() = physics + reward; train() = PPO; main() = experiment + graphs.

SAY “The program builds the joint, trains two policies, evaluates both, and saves the graphs.”

Does the joint follow the target?

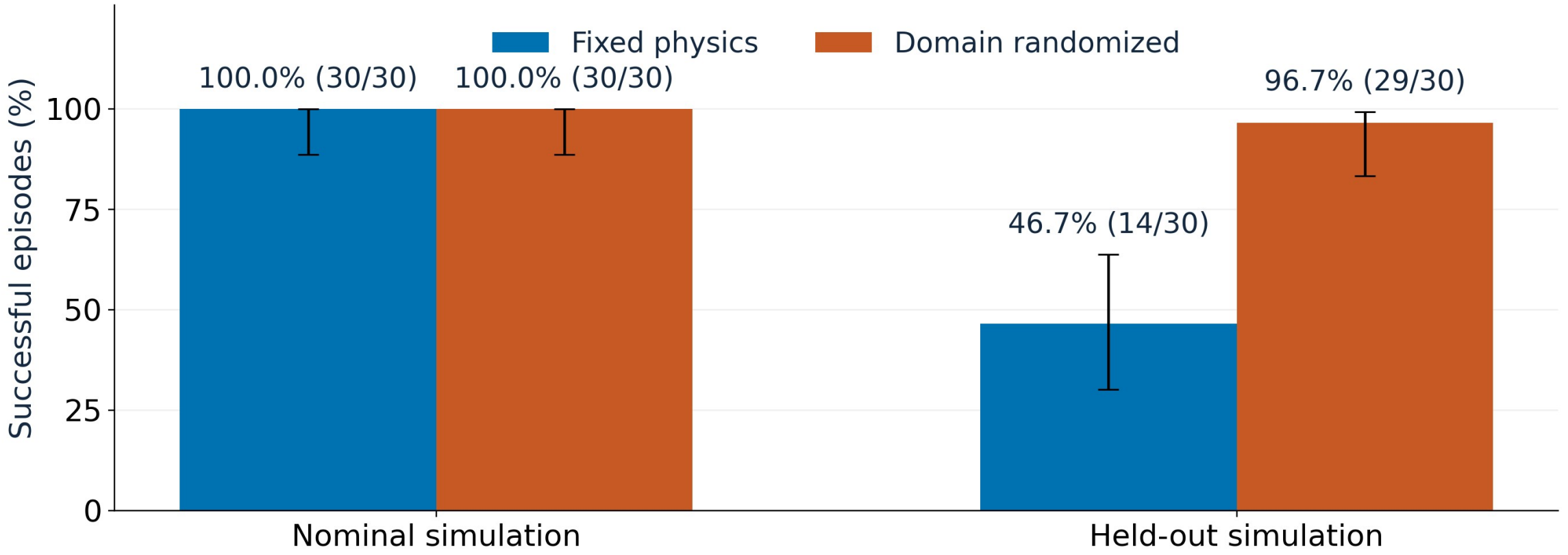


Dashed = desired angle • Colored = actual angle • Same held-out trial for both policies

Seed 1000: both pass RMSE < 0.32 rad. Overall success uses all 30 trials, shown next.

SAY “The dashed line is the desired angle; the colored line shows what the joint does.”

What do the results mean?



Held-out mean RMSE: 0.3299 rad fixed → 0.2250 rad randomized. Both reach 100% nominal success.

SAY “Randomization improved held-out performance for this policy pair; both policies succeeded in nominal simulation.”

How does this connect to Reading 5?

TAN ET AL. (2018)

Learned control, actuation, and randomized simulation physics.

ZHAO ET AL. · SURVEY

The reality gap: simulation and real hardware behave differently.

SUTTON & BARTO · CH. 13

Policy gradients are the foundation.
PPO: a clipped actor–critic method.

MUJOCO

Richer multibody and contact physics.
This example uses a simpler joint model.

SAY “The readings connect randomized simulation, policy learning, and the gap between a model and a robot.”

A real robotics application and the takeaway

Application: a quadruped joint must track commands as its payload, friction, and communication delay change.

PAYLOAD

Changes joint inertia

WEAR

Changes friction

ACTUATION

Introduces command delay

This run: randomized training improved held-out tracking and success.

Go2 / ANYmal inspiration • one joint • one training seed • simulation only

SAY “Training across plausible variation may help controllers tolerate changing loads, friction, and actuator delays.”